# SHA-256 Preimage Bench

## Direct recovery of one 32-bit message word

Rounds 1–16 of a fixed single-block message family

# The recovery problem

# 40 bytes

**4 bytes vary. The other 36 are fixed.**

Standard padding produces a single 512-bit block.

The input to recovery is the full 256-bit output and the round count.

```
Message = W0 || 01 02 … 24 (hex)
```

# Why the method stops at round 17

## Rounds 1–16

Undo feed-forward and reverse later rounds using fixed words. Solve for W0 in round 1.

## Round 17

The first expanded word depends on W0.

$$W16 = W0 + \sigma1(W14) + W9 + \sigma0(W1) \mod 2^{32}$$

# 1,024 verified full-width recoveries

64 seeded 32-bit inputs at each of 16 round counts

## Timing on a separate 12-bit input set

| Rounds | Direct recovery | Ascending enumeration |
|---|---|---|
| **1** | 8.30 µs | 4.83 ms |
| **4** | 16.60 µs | 10.53 ms |
| **8** | 25.85 µs | 18.12 ms |
| **16** | 46.70 µs | 33.57 ms |

Medians of 24 paired trials per row. Both methods stop at a verified match.

# Earlier Z3 solver comparison

A fixed 12-bit input space with a 2-second budget per solver check

| Rounds | Bit-vector | Bit-vector + GF(2) |
|---|---|---|
| **4** | Verified witness | Verified witness |
| **8** | Verified witness | Verified witness |
| **16** | Unknown | Unknown |
| **64** | Unknown | Unknown |

At 16 and 64 rounds, both solver checks ended without a determination.

# Reproducing the experiment

Python 3.11 or newer

```
python -X utf8 -m unittest discover -s tests -v
python -X utf8 benchmark.py
```

The tests check the 64-round digest against hashlib.

The benchmark exports the seed and every trial to JSON.